Your Paper Title

Anonymous Author(s)

Abstract

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Keywords

large language models, multi-agent systems, software engineering, debugging, requirements engineering

1 Introduction

Recent studies have increasingly emphasized the central role of requirements in AI-driven software engineering. Robinson et al. [3] envision a future paradigm in which end-users can directly drive the entire software development lifecycle through multimodal requirements expressed in natural language, images, audio, and video, highlighting requirements as the core interface between humans and generative software systems. Along this direction, Han et al. [2] proposed ARCHCODE, a requirement-aware code generation framework that explicitly extracts and extrapolates both functional and non-functional requirements from textual descriptions to guide code and test generation, demonstrating that requirement-centric reasoning can substantially improve code correctness and requirement satisfaction. Meanwhile, Darif et al. [1] systematically reviewed Controlled Natural Language (CNL) approaches for requirements specification and showed that reducing ambiguity while preserving stakeholder readability remains a longstanding challenge in requirements engineering. Their findings further reveal limitations in automated tooling, domain vocabulary integration, and practical adoption, suggesting that intelligent and cognitively guided requirement understanding remains an open research problem. Together, these studies indicate a growing shift from syntax-oriented software generation toward requirement-centric and cognition-aware software engineering paradigms.

2 Related Work

In this section, we review the existing literature on multi-hop question answering (MHQA) and related areas. We categorize the related work into three main groups: (1) traditional MHQA approaches, (2) neural network-based methods, and (3) transformer-based models. For each category, we discuss the key contributions, strengths, and limitations of the approaches. We also highlight how our proposed method differs from and improves upon these existing works. Finally, we identify the gaps in the current research that our method aims to fill.

3 Methodology

In this section, we describe the methodology of our proposed approach for multi-hop question answering (MHQA). We first provide an overview of the architecture of our model, including the key components and their interactions. We then detail the training procedure, including the loss function and optimization strategy. Finally, we discuss any additional techniques or modifications that we applied to enhance the performance of our model, such as data augmentation or regularization methods.

Figure 1: A sample figure.

4 Experiments

In this section, we present the experimental setup and results of our proposed method for multi-hop question answering (MHQA). We first describe the datasets used for evaluation, including their characteristics and the metrics employed to assess performance. We then compare our method against several baseline approaches, highlighting the improvements achieved. Finally, we conduct an ablation study to analyze the contribution of different components of our model and discuss the implications of our findings.

5 Threats to Validity

In this section, we discuss the potential threats to the validity of our study and the measures we have taken to mitigate them. We categorize the threats into three main types: (1) internal validity, (2) external validity, and (3) construct validity. For each type, we identify specific threats that could affect the reliability and generalizability of our findings. We also describe the steps we have taken to address these threats, such as using a rigorous experimental design, ensuring a representative sample, and employing appropriate statistical analyses. Finally, we acknowledge any remaining limitations and suggest directions for future research to further validate our results.

6 Conclusion

In this paper, we have presented a novel approach to multi-hop question answering (MHQA) that addresses the challenges of reasoning across multiple pieces of information. Our method leverages a combination of traditional MHQA techniques, neural network-based methods, and transformer-based models to achieve state-of-the-art performance on several benchmark datasets. Through extensive experiments, we have demonstrated the effectiveness of our approach compared to existing methods, and our ablation study has highlighted the importance of each component of our model. We believe that our contributions provide valuable insights into the design of MHQA systems and pave the way for future research in this area. Future work could explore further enhancements to our model, such as incorporating additional contextual information or exploring alternative architectures for reasoning. Additionally, we plan to investigate the application of our method to other related tasks in natural language processing, such as reading comprehension and

information retrieval. Overall, our work represents a significant step forward in the field of multi-hop question answering and has the potential to inspire further advancements in this exciting area of research.

References

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[3] Diana Robinson, Christian Cabrera, Andrew D. Gordon, Neil D. Lawrence, and Lars Mennen. 2025. Requirements Are All You Need: The Final Frontier for End-User Software Engineering. *ACM Trans. Softw. Eng. Methodol.* 34, 5 (May 2025), 141:1–141:22. doi:10.1145/3708524

A Appendix

In this section, we provide additional materials and details that support the main content of the paper. This may include supplementary figures, tables, or explanations that are not essential to the main narrative but provide valuable insights or additional information for readers who are interested in a deeper understanding of our work. We also include any relevant code snippets, data preprocessing steps, or hyperparameter settings that were used in our experiments. Finally, we may include any additional analyses or results that were not included in the main body of the paper but are relevant to our research.